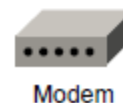
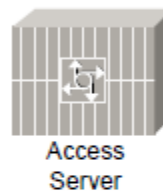
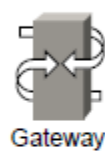
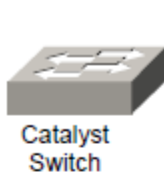


-FCI - 2026-2027
Lectures Notes on Networks

Part1

Network Fundamentals

Network Device Symbols



Communicating in a Network-Centric World

Humans depend on the interaction with others for daily needs. Throughout human history, people, with few exceptions, have depended on the structure of various community networks for safety, food, and companionship. People have been networking for a very long time.

Now, the Internet allows people to instantaneously share all types of communication—documents, pictures, sound, and video—with thousands of people near and far away using compute

Communication: An Essential Part of Our Lives

Communication in our daily lives takes many forms and occurs in many environments. We have different expectations depending on whether we are chatting through the Internet or participating in a job interview. Each situation has its corresponding expected behaviors and styles.

These expectations are the rules of communication, and some of the elements are universal.

Taking a closer look at the way humans communicate will introduce many of the necessary elements of network communication as well.

The Network as a Platform (Communicating over Networks)

- The ability to reliably communicate to anyone, anywhere, is becoming increasingly important to our personal and business lives. Adding to the demand of immediacy is the requirement that different types of messages, such as phone, text, and video, be accommodated as normal forms of communication. To support the immediate delivery of the millions of various messages being exchanged among people all over the world, we rely on a web of interconnected networks. The following sections describe communication over networks, different elements that make up a network, and convergence.
- Networks directly impact how we live, and the role of networks is increasingly important to people in all parts of the world. The task of reliably delivering millions of messages simultaneously would be too much for any one network to perform. Therefore, a web of smaller, interconnected networks of various sizes and capabilities delivers the many messages and data streams around the world.

A network is a set of devices (often referred to as nodes) connected by communication links.

- A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.

“Computer network” is a collection of autonomous computers interconnected by a single technology. Two computers are said to be interconnected if they are able to exchange information.

- Networks come in many sizes, shapes and forms, as we will see later.
- They are usually connected together to make larger networks, with the Internet being the most well-known example of a network of networks.

Why Networks? (Benefits of Network) :

1. Data Exchange
2. Sharing Resources

-Data exchange in a network is the process of sharing and transferring data between two or more connected systems.

Examples : email, VoIP, and e-commerce .

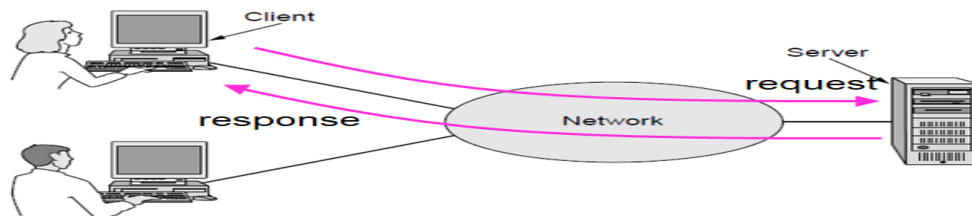
-Sharing Resources :

Networks allow users to share hardware, software, and other resources.

Examples: Printers - Storage - Servers - Applications -Internet connections

Example: Several computers can share one network printer.

Companies use networks and computers for resource sharing with the client-server model:



Network Tasks:

- Network tasks are the fundamental functions performed by a network to facilitate communication, data transfer, resource access, and the secure and reliable operation of interconnected devices.
- In other words, they are the functions that a network performs to enable communication between devices and users. A network performs tasks such as:
 1. Connecting devices (Communication)
 2. Transmitting data
 3. Addressing devices
 4. Routing data
 5. Forwarding packets
 6. Providing access to resources and services
 7. Controlling access
 8. Protecting data and communications
 9. Managing network traffic
 10. Monitoring and maintaining network operation

1. Communication

The primary task of a network is to enable devices and users to communicate and exchange information.

Part1

Network Fundamentals

Examples: Data communication - Email - Voice communication - Video conferencing

2. Data Transmission

The network transports data from a source to a destination.

Sender → Network → Receiver

3. Addressing

The network identifies devices and provides a mechanism for determining the source and destination of data.

Examples: MAC addresses - IP addresses

4. Routing

The network determines the appropriate path for packets to travel from the source to the destination.

5. Switching and Forwarding

Network devices such as switches and routers forward data toward their intended destination.

6. Network Management

Networks must be monitored, configured, maintained, and troubleshot.

7. Access Control

The network controls which users and devices are permitted to access resources and services.

8. Security

The network helps protect data and resources against unauthorized access and attacks.

Examples: Authentication - Encryption - Firewalls

9. Error Control and Reliability

Networks provide mechanisms to detect, recover from, or minimize communication errors and service failures.

10. Traffic Management

Networks manage traffic to maintain acceptable performance and avoid congestion.

Examples: Traffic prioritization - QoS - Congestion control - Traffic shaping

A useful distinction between Network benefits and Network tasks:

Network Tasks = What the network does.

Network Benefits = What we gain from using the network.

Example:

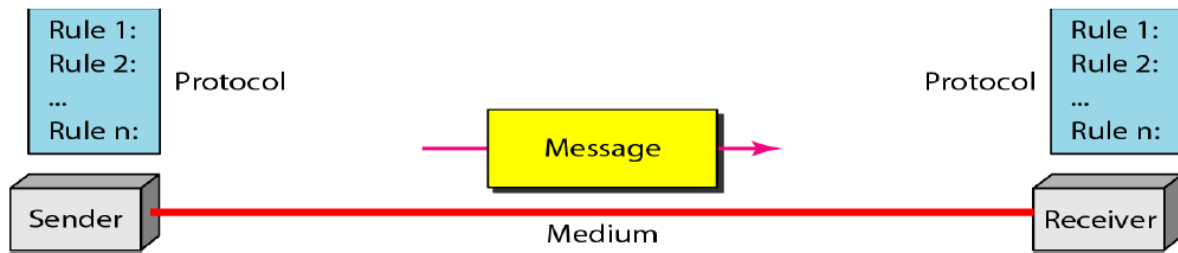
Task: The network transmits data between two computers.

↓

Benefit: Users can communicate and share information.

So, the task is the technical function, while the benefit is the resulting advantage.

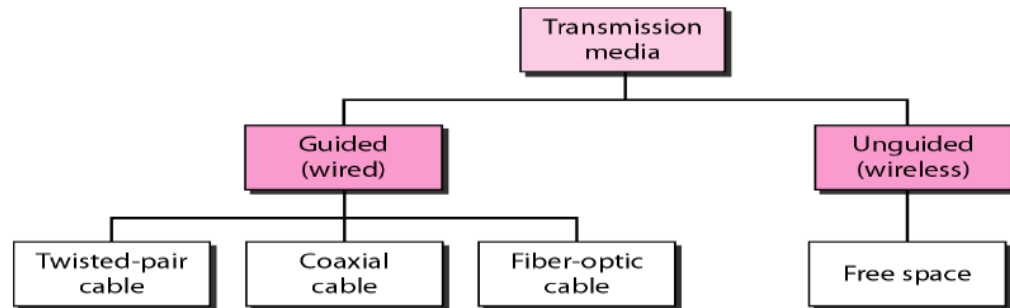
Components of Network (The Elements of a Network):



- ✓ **Devices** : They are used to communicate with one another
They send and receive the data message. They can be a computer, workstation, telephone handset, video camera, and so on.
- ✓ **Medium**: The transmission medium is the physical path by which a message travels from sender to receiver. The main two types of transmission media are wired (guided) and wireless (unguided) .
- ✓ **Messages** Information that travels over the medium. The message is the information (data) to be communicated. Information includes text, numbers, pictures, audio, and video.
- ✓ **Protocols** or Rules that Governs how messages flow across network.
It represents an agreement between the communicating devices. Without a protocol, two devices may be connected but not communicating, just as a person speaking French cannot be understood by a person who speaks only Japanese.
- ✓ **Intermediary (Networking) Devices**.
Intermediary devices connect the individual hosts to the network and can connect multiple individual networks to form an internetwork.
Some work inside the LAN performing switching functions, and others help route messages between networks. The Below Table lists some intermediary devices and their functions.

Device Type	Description
Network access devices	Connect end users to their network. Examples are hubs, switches, and wireless access points.
Internetwork devices	Connect one network to one or more other networks. Routers are the main example.
Communication servers	Route services such as IPTV and wireless broadband.
Modems	Connect users to servers and networks through telephone or cable.
Security devices	Secure the network with devices such as firewalls that analyze traffic exiting and entering networks.

Classes of transmission media

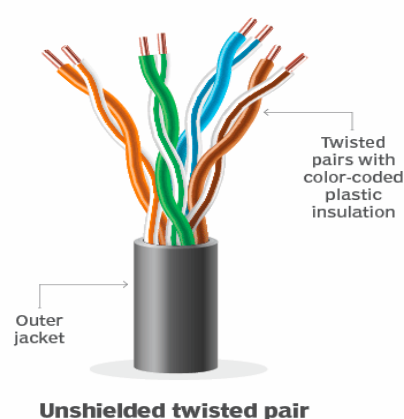
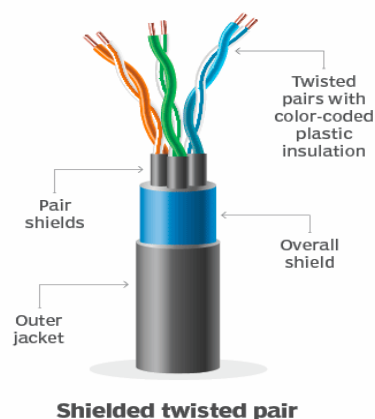


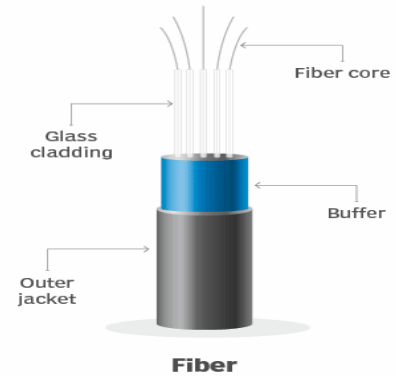
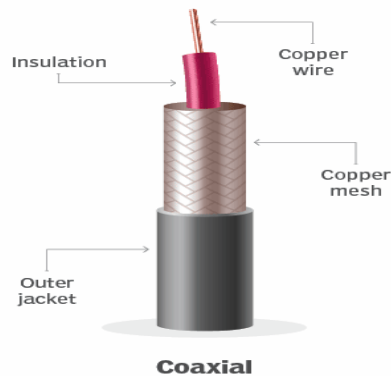
Guided (Wired) Media

Guided media, which are those that provide a medium from one device to another, include twisted-pair cable, coaxial cable, and fiber-optic cable.

The main types of network cables are :

- ❖ **Twisted pair cables** (UTP and STP) are widely used in Ethernet LANs, with UTP being the most common and STP providing better noise resistance.
- ❖ **Coaxial cables**, unshielded twisted pair (UTP), shielded twisted pair (STP). They are used for transmitting high-frequency signals.
- ❖ **Fiber optic cables** that are reserved for high-speed, long-distance applications due to its superior speed and bandwidth.
- The other is wireless or unguided transmission medium (e.g.: Infrared, Radio link, Microwave link, Satellite link, Bluetooth, WIFI)





Wireless Transmission

- Wireless transmission media are communication channels that transmit data **without physical cables**, using electromagnetic waves through air or space.
- Instead of copper wires or fiber optics, signals travel as waves.
- Wireless transmission uses **electromagnetic signals** to carry information between sender and receiver through:
 - Air
 - Vacuum (space)
 - Water (in special cases)
- Examples of wireless Transmissions are Bluetooth , Wi-Fi , Microwave , Mobile and Satellite communications.

Basic Definitions

Data Rate is the speed at which data is transferred within path or channel. Or the amount of data transmitted per second over a communication channel.

It includes useful data + overhead bits (headers, error correction, etc.).

Example: If a link transmits 1 Mbps of total data (including headers), the data rate = 1 Mbps.

Bit Rate: It is the number of bits transmitted or received per second. Often used as a synonym for data rate.

Sometimes refers to the raw transmission speed before encoding or redundancy is added.

Information Rate: it is the rate of useful information only (excluding overhead, redundancy, or control bits).

-FCI - 2026-2027

Lectures Notes on Networks

Part1

Network Fundamentals

It represents the net information flow that carries meaning. It is Always less than or equal to the data rate.

Example: If Data rate = 100 kbps and Overhead = 20% Then $\Rightarrow$ Information rate = 80 kbps

Link Rate: It is the theoretical maximum speed of a physical link between two devices (e.g., Wi-Fi, Ethernet). It is determined by the technology or standard being used.

Example: Wi-Fi 6 $\rightarrow$ link rate = up to 1200 Mbps. Ethernet 100Base-T $\rightarrow$ link rate = 100 Mbps.

Throughput: The actual rate of successful data delivery over a network or communication channel.

It measures how much useful data actually reaches the destination per second, including all real-world effects such as delays, congestion, and retransmissions.

It depends on: Network congestion, Signal quality (errors and retransmissions) , Protocol overhead and Processing delays.

Relationship Summary: $\text{Throughput} \leq \text{Information Rate} \leq \text{Data Rate} \approx \text{Bit Rate} \leq \text{Link Rate}$.

In short:

- **Bit rate and data rate** describe how fast bits are sent.
- **Information rate** describes how much meaning those bits carry.
- **Link rate** : describes maximum speed of a physical
- **Throughput** describes how much useful data actually arrives.

Example Suppose:

Term	Meaning	Includes Overhead	Real or Theoretical?	Example
Bit Rate	Number of bits transmitted per second	Yes	Theoretical	100 Mbps
Data Rate	Total data sent (useful + control bits)	Yes	Theoretical	100 Mbps
Link Rate	Maximum speed of the physical link	Yes	Theoretical	100 Mbps
Information Rate	Useful information bits only	No	Theoretical	80 Mbps
Throughput	Actual data successfully delivered	No	Real (measured)	60 Mbps

Link Rate = 200 Mbps (theoretical maximum) ,

Data Rate = 100 Mbps (raw transmission rate)

Information Rate = 80 Mbps (useful bits after overhead)

Throughput = 60 Mbps (what the receiver actually gets, due to network delays, interference, etc.)

Part1

Network Fundamentals

Network Hardware (Network Core and Network Edge)

These are two fundamental parts of a network architecture, especially important when studying WANs.

1. Network Edge

Network Edge is the part of the network closest to end users and end devices.

Typical Edge Components: PCs and laptops - Smartphones -IP phones - IoT devices - Access points
Access switches -Edge routers

Main Functions

The Network Edge is responsible for:

- Connecting end devices to the network
- Sending and receiving user traffic
- Applying access and security policies
- Providing network access to users and applications

2. Network Core

Network Core is the high-speed central part of a network that transports traffic between different networks, locations, or services. It is sometimes described as the backbone of the network.

Typical Core Components

High-speed routers - Core switches - High-capacity links -Redundant connections

Main Functions

The Network Core primarily focuses on transporting data quickly and reliably between different parts of the network. Its features are :

- Fast packet forwarding
- High availability
- High capacity
- Low latency
- Reliability
- Traffic transport between network segments

Simply,

Network Edge = Connects users and networks

Network Core = Transports traffic between networks.

